
# Lesson 4

*ILAI (M1) @ LAAI I.C. @ LM AI*

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*These slides draw very heavily from Simone Martini's slides.*

# One-slide recap of Lecture 2

Function: sequence of commands with a name. Returns a value.

Defined with `def`, name, parenthesis, formal parameters (names)

Called with name and actual parameters (*arguments*, are expressions)

Value of actual parameters bound to the formal parameters, which are local names

Also any name occurring to the left of an assignment is local to the function

Function without return, return the value `None` (of type `NoneType`): a value of expressions for which the value is irrelevant (we are interested in «side effects», actions on the internal state).

We can import names (and functions) from libraries with `import` or `from … import`

We can use `for <name> in <sequence>`: to scan every element of a sequence. `<name>` is just a name like all the others (no local scope).

The «`in`» in the `for` should not be confused with the boolean operator `in`, which returns `True` if an element (for strings only, also substrings) appear in a sequence.


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# One-slide recap of Lecture 3

Object: value enveloped in an identity. Different objects (identities) may have the same value
Identity accessible with `id()`. Identities compared with `is`. NB: `==` compares values, instead

Objects with same type have same methods. Methods are invoked on objects with the dot `.`

`tuple` is a compound/structured type (group of values each of its type). Immutable sequence.
Same operations of other imutable sequences (for, concatenation, access with index…)

Generalized assignment: <sequence of # names> = <sequence with the same # of elements>

Slice: a «window» on a sequence. `[b:e:p]` between `b` included and `e` excluded, with step `p` (default: 1. Can be negative to select backward)

Operations create new objects. On the contrary, assignments can create aliasing situations

`range`: a special sequence (not expanded) of integers. Indexes work like in slices. `for i in range()` … to iterate over numbers – usually indexes of a sequence


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# Unbounded iteration

Problem:

We randomly extract integers from 0 to 99

Given an integer x (between 0 and 99)

We want to count how many extractions we perform before extracting x


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# Unbounded iteration

Problem:

We randomly extract integers from 0 to 9 and count

Given an integer x (between 0 and 9)

We wanto to count how many extractions we perform before extracting x

We don't know *beforehands* how many extractions!

bounded iteration is not the right construct!

file: while.py


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# while command
# Unbounded iteration

```
while < bool-expr> :
    <block>
```

Semantics:

1. Evaluate < bool-expr> (the *guard*)

2.1 If the guard is False, terminate the command (i.e., proceed with the command following the block, at the indentation of «while») ("*we exit from while*")

2.2 If the guard is True, evaluate <block> (the «while» *body*) ("*we enter inside the while* ")

3. Go back to step (1)

# *Bounded* (`for`) iteration: ingredients

```
for i in sequence:
    <body>
```

1. What to repeat: <body>
2. Initial value for `i` : first element of `sequence` (if any)
3. How to pass to next iteration: `i` takes *next* element of `sequence`
4. Termination: `sequence` is exausted

All these ingredients *must be given explicitely*
in an unbounded iteration (`while`)

# *unbounded* (`while`) iteration: ingredients

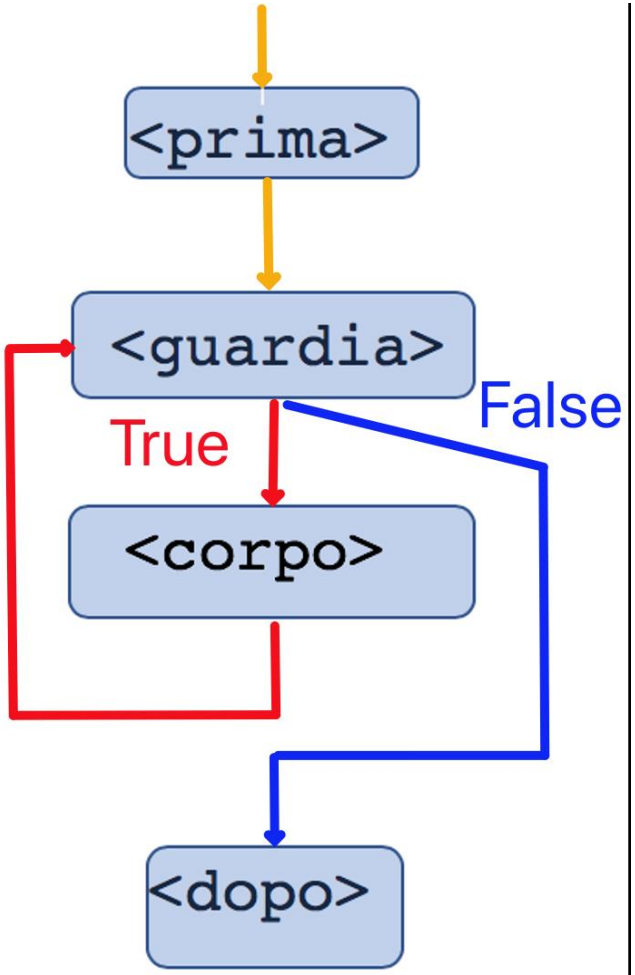
```
<before>
while  <guard> :
    <body>
```

1. What to repeat: <body>
2. Initial value for the guard : ➔ *before the while*
3. How to pass to next iteration: ➔ *inside the <body> the guard may (should) change*
4. Termination: ➔ *<guard>*

All these ingredients *must be given explicitely*
in an unbounded iteration (`while`)

# while

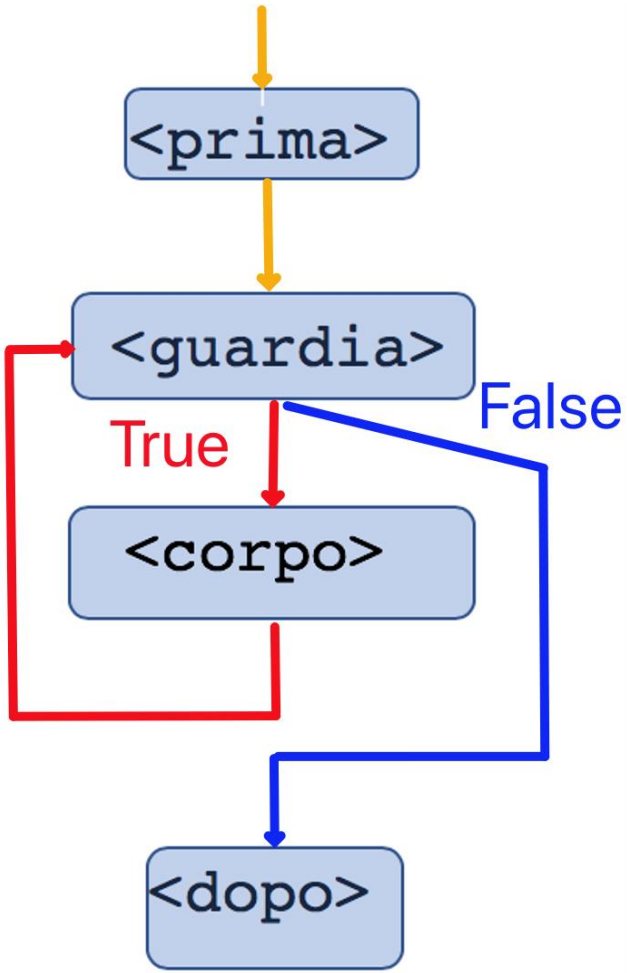
```
<before>
while <guard>:
    <body>
<after>
```



# while

```
<before>
while <guard>:
    <body>
<after>
```

1. <before> must guarantee an initial value to <guard>
2. <body> must modify the value of <guard>



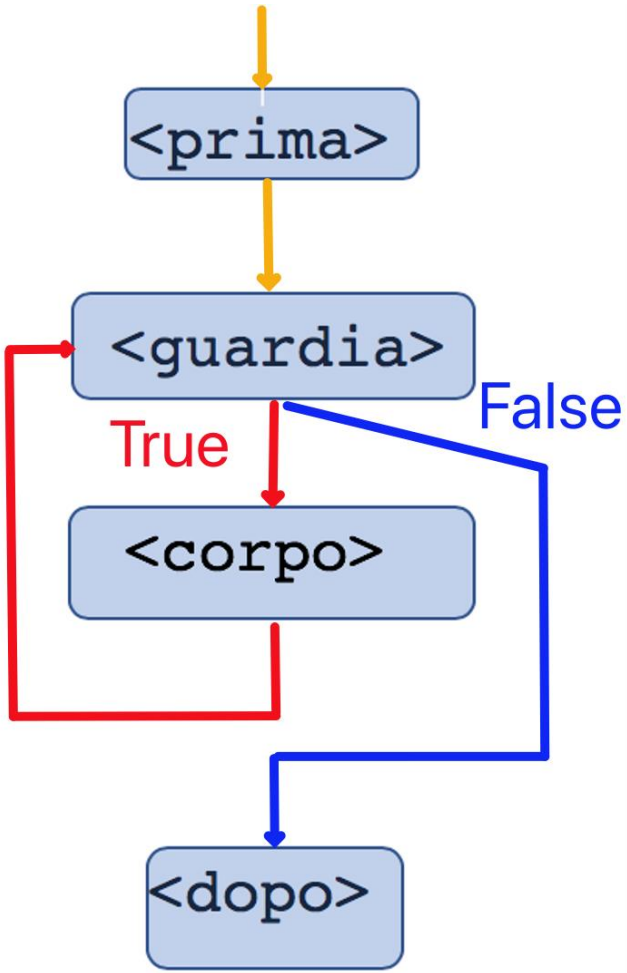
# while: programming

```
<before>
while <guard>:
    <body>
<after>
```

A "while" command is never alone

- before initializations
- body must modify the guard

They are *not* grammatical requirements:
good programming practice



# for expressed through while

```
for i in range(len(s)):
    <body>
```

With some approximation
corresponds to:

# for expressed through while

```
for i in range(len(s)):
    <body>
```

With some approximation corresponds to:

```
i=0
while i < len(s):
    <body>
    i = i+1
```

It does not"freez" neither s nor its lenght

Unlike for, Python's while is very similar to many other languages ☺

## But compare:

```
sum=0
for e in s:
    sum=sum+e
print(sum)
```

```
sum=0
for i in range(len(s)):
    sum=sum+s[i]
print(sum)
```

```
i=0
sum=0
while i < len(s):
    sum=sum+s[i]
    i = i+1
print(sum)
```

**Every time I can**, I use:

- for without indexes (directly on the sequence)
- for instead of while

# Is it a correct program?

```
while 1==1:
    x=100
print(x)
```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# Is it a correct program?

```
while 1==1:
    x=100
print(x)
```

*Syntactically* correct

*Semantically* wrong: it is a infinite loop which never interacts with the external world

# A new data type:

# lists


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# Lists: `list`

- structured type (like tuple)

- *mutable* sequences of objects

# Lists

Values: finite sequences of values, mutable

Presentation: [<object1>, … , <objectk>]

[] is the empty list

a list with a single element: [<element>]

(or also [<element>,] )

Operations: concatenation (+), repetition (*), length len(), selection […]

but also…

# Mutable

Contrary to all types we have seen so far:

we may modify the *value* of an object

preserving its identity

Example:

```
L=[10,20,30]
L[1] = 200
```

The object bound to L did not change

Its value changed

Warning: LHS of = is *not* a name

# Assignment, again

The simplest form (from L1):

<name> = <expression>

Now:

<selection> = <expression>

<selection> must be on a mutable object (and cannot be a name)

Plus the extended versions we saw in L3:

<name1>,…,<namek>=<sequence with exactly k elements>

A,B = B,A

L[i+1],L[i]=L[i],L[i+1]

# Assignment, again

Now:

<selection> = <expression>

<selection> must be on a mutable object (and cannot be a name)

Semantics:

- evaluate <*expression*> and obtain an object *o*
- evaluate <selection> and obtain a component *p* of a modifiable object *m*
- the component *p* of *m* is modified in *o*

Observe: there is *no* modification of the bindings between names and values in the internal state


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# Function that takes a list of int and modifies every 0 into 100


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# State evolution

```
L = [1, 2]
V = [10, 20]
W = L + V
L = L + [3]
L[0] = 100
Z = ['s', V, 10]
Z[1][0] = 0
print(V)
```

# State evolution

```
L = [1, 2]
V = [10, L, 20]
W = [30, L, 40]
printV)
print(W)

LL = L[:]
VV = V[:]
z = …
```

# Example: double the value of elements

Given a list L of int, modify any element, transforming it in its double

=> file double.py

given L=[10,20,30,40],

transform L in [20,40,60,80]


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# Aliasing and side effects

```
L = [10,20,30]
V = L
L[0]=100
print(V)
```

Since there is aliasing (V and L are alias for the same object),
V has been modified by *side effect*

We also say:
`L[0]=100`
has the side effect to modify `V`

# Aliasing and side effects, 2

We may have aliasing also on tuples (or int, or float etc.):

```
T = (10,20,30)
R = (100, T, 300)
```

the “same” on lists:

```
L = [10,20,30]
W = [100, L, 300]
```

On list we may modify L, and this will be seen also from W

# Aliasing and side effects

Small examples on

aliasing_example.py


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# Aliasing and side effects, moral

Aliasing and side effects are present in the language *by design*

They are important to do meaningful things!

Yet, they are a delicate, subtle ingredient of the language

- to be understood
- to be controlled
- because they are source of errors which are difficult to locate and discover

# Deleting elements from a list: `del`

Command `del`:

```
del L[i]
```

Semantics:

L[i] is eliminated from L

the elements "following" L[i] are "shifted to the left"

(indexes are recomputed)

# Given list L of int, delete the zeros

```
L = [1,2,0,0,3,0,5,0]
```

# Given list L of int, delete the zeros

```
L = [1,2,0,0,3,0,5,0]
i=0
while i<len(L):
    if L[i]==0:
        del L[i]
    else:
        i=i+1
# here L is [1,2,3,5]
```

Observe that the index is incremented *only* when `del` is *not executed* (file removezero.py)

# It does not work with for!

```
L = [1,2,0,3,0,5,0]
i=0
while i<len(L):
    if L[i]==0:
        del L[i]
    else:
        i=i+1
# here L is [1,2,3,5]
```

```
L = [1,2,0,3,0,5,0]
for i in range(len(L)):
    if L[i]==0:
        del L[i]
# here L is ??
```

WRONG: IT DOES NOT WORK!

# Don't use for on lists in presence of side effects!

Never (never !)

use `for` on lists,

*if their length*

*is modified in the `for` body*

# Don't use for on lists in presence of side effects!

*The heading of the `for` command freezes the sequence object (its identity)*

*not its value*


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# del

In the "modifying assignment"

<selection> = <expression>

<selection> *must be an explicit selection*

In

del <selection>

<selection> *must be an explicit selection*

Compare:

```
L=[10,20,30]
del L[1]
```

```
L=[10,20,30]
e=L[1]
del e
```

Beware: del *<name>* will remove *<name>* from the internal state!

# Add/delete elements: methods

Invoking a method on an object of type list

may *modify the object*


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# Some methods on list

L.append(x) L[len(L):]=[x]

append x as last element of L; returns None

L.insert(i,x) L[i:i]=[x]

insert x in L at index i (shifting indexes),
or append if i>=len(L); returns None

L.extend(V) L[len(L):]=V

extends L with list V (in place concatenation)

L.clear() del L[:]

empties L (mantains the identity)

➔➔➔ all of them return **`None`**

# Add/delete elements: slice

<selection> = <expression>

<selection> may be a *slice*

*If <selection> is a slice:*

- *RHS must be a list, whose elements are inserted for the slice*
- *as a result of the assignment, the list at LHS may grow, or shrink*

# Add/delete elements: slice

Selection may be a *slice. Examples:*

```
L=[1,2,3]
```


```
L[0:2]=[10,20]
```


```
L[0:1]=[5,6]
```


```
L[-1:]=[4]
```


```
L[1:1]=[1,2]
```


```
L[0:2]=[]
```


```
L[0:0]=[1]
```


```
L[len(L):]=[30]
```

# Warnings

1)

del L[i:j]  is equivalent to  L[i:j]=[]

2)

L[1:2]=[4,5,6] and L[1]=[4,5,6]

*are two (very!) different things*

- L[1:2]=[4,5,6] *extends* L

- L[1]=[4,5,6] leaves unchanged the length of L,: the element L[1] is replaced by the (list) object [1,2,3]

# Warnings

Semantically equivalent to slicing

- In general, using methods is more efficent

# Careful

L=L+[elemento]

is not the same as

L.append(elemento)

In the first, a new object is created
In the second, no new object

Then…

# Don't use for on lists in presence of side effects!

The heading of the `for` command freezes the sequence object (its identity)

*not its value*

Compare:

```
L=[10]
for e in L:
    L.append(e)
print(L)
```

```
L=[10]
for e in L:
    L=L+[e]
print(L)
```

# Warning!

The heading of the `for` command freezes the sequence object (its identity)

*not its value*

Compare:

```
L=[10]
for e in L:
    L.append(e)
print(L)
```

```
L=[10]
for e in L:
    L=L+[e]
print(L)
```

The first is an infinite loop, the second prints [10,10]

# Careful

L=L+[elemento]

is not the same as

L.append(elemento)

Suppose there is aliasing on the object bound to L:

```
L=[10,20,30]
W=L
L.append(40)
print(W)
L=L+[60]
print(W)
print(L)
```

# One example with methods

Read from the keyboard a series of positive integers, until the user inserts 9999
Store the numbers in a list

file inser.py

# assignment: one more variant

*augmented assignment*

In iteration scans one often uses:

```
res = res + val
```

or also

```
count = count*val
```

We may use *contracted forms*

# assignment: one more variant

*augmented assignment*

In iteration scans one often uses:

```
res = res + val
```

or also

```
count = count*val
```

We may use *contracted forms*

```
res += val

count *= val
```

# Augmented assignments

+=  -=  *=  /=  //=  %=  **=

and even others

Operations are overloaded:

T += v

is addition, or concatenation, etc.

depending from the type of T

# Augmented assignments

`x+=v` `x=x+v`

Similar, *but not the same*, to its "normal" analogue

1. the operation is done *in place* if possible
2. x *is evaluated only once*
3. *LHS is evaluated first*

*Let's dig deeper*

# Augmented assignments

Similar, but *not the same* as their "normal" analogues

1. The operation is done *in place* if possible

```
L = [1,2,3]
L +=  [4]

V = [1,2,3]
V = V + [4]
```

are two *different things:*

+= is extend (append) on lists!

# Augmented assignments

Similar, but *not the same* as their "normal" analogues

2. x *is evaluated only once*

Let f(x) be a function

```
L[f(i)] +=  val
```

```
L[f(i)] = L[f(i)] + val
```

+= calls f(i) only once

➔ if f() has side effects, they happen once only!

# Augmented assignments

Similar, but *not the same* as their "normal" analogues

*3. LHS is evaluated first,* then RHS, then the binding in the state is modified

Let f(x) be a function

```
L[f(i)] +=  exp
```

*If f(i) has side effects, they happen before the evaluation of exp*

(file augmented.py)

# More methods on list

L.remove(x)

remove from L the first element L[i] equal (==) to x
error if `x not in L`
returns `None`

L.copy() L[:]

returns a (shallow) copy of L

L.reverse()

reverses L (in place)
returns None
What is the difference with L=L[::-1] ??
remember the mantra: "operations create new objects"

L.sort()

sort L (in place)
returns None
compare with the builtin function M = sorted(L)
remember the mantra: "operations create new objects"

# More methods on list

L.pop(i)

returns and removes the element at index i

pop() returns and removes L[-1]

returns a value: L[i]
modify in place the list, removing the element at index i
leave unchanged the identity of L

# Can functions modify global mutable objects?

# Can functions modify global objects?

Observe that with *immutable* objects, this is impossible:

```
b=20
def f(a):
    a=10
    b=100
    return None
f(b)
print(b)
```

a,b in the body of f are local names!

# Intermezzo: global (more in next lecture)

Observe that with *immutable* objects, this is impossible,

unless a global declaration is used

```
b=20
def f(a):
    global b
    a=10
    b=100
    return None
f(b)
print(b)
```

a is local; b is the one in the main frame

# Can functions modify global mutable objects?

With mutable objects: possibile, in several ways

1. *Through a mutable actual parameter:*

```
b=[20]
def f(a):
    a[0]=100
    # return None
f(b)
print(b)
```

In Pythontutor…

Of course the assignment `a[0]=100` would results in an error when f is called on an immutable actual parameter

# Can functions modify global mutable objects?

With mutable objects: possibile, in several ways

2. *Through a global name, bound to a mutable object*

```
b=[20]
def f():
    b[0]=100
f()
print(b)
```

In Pythontutor…

Observe: at the left of = there is a *selection* not a name

Hence the rule: "name at the left of = is local" cannot be applied

# Double (nested) iteration

Sometimes we need to iterate *inside* another iteration.
This programming schema is called double (or nested) iteration

Trivial example: Given a tuple of tuples of integers, count the zeros

```
def zeros(ToT):
    count=0
    for t in ToT:
        for e in t:
            if e==0:
                count += 1
    return count
```

file tuples.py

# Some exercises and challanges


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# For home 1: ordered insertion

Write a function

```
def ins_ord(L,el):
    '''L: ordered list, in increasing order
insert el in L, in place'''
```

Example:

S=[2,4,6,8]

ins_ord(S,5) must modify S in [2,4,5,6,8]

Some solutions in file inserimento-in-lista-ord.py

# For home 2: Floyd triangles

A *Floyd triangle* of order 5:

five rows, with the integers from 1; any row has one more element of the previous row

```
 1
 2  3
 4  5  6
 7  8  9 10
11 12 13 14 15
```

Write a function Floyd(n) which prints a Floyd triangle of order n

We want a *pretty print*

# For home 2: Floyd triangles

We want a *pretty print*

print may have additional arguments "by keyword":

`sep`: string used to separate values *in the same print* (default: ' ')

`end`: string used to *terminate, after a print* (default: newline: '\n')

```
print(10,20)
print(30,40)
print(10,20,sep=' & ',end=' * ')
print(30,40,sep=' & ')
print(50)
```

```
10 20
30 40
10 & 20 * 30 & 40
50
```

\t is "tab"
\n is "neline"
etc: look in the doc

# Double iteration: matrices

Python does not have a type for array/matrices

**(but we will see and use the library NumPy)**

AS TOY EXERCISE for nested iterations, encode a *n*x*m* matrix as:

- list of lists
  - a list of *n* elements (the rows),
  - each element is a a list of m elements (the *m* elements of the row)

$$A = \begin{pmatrix} 0 & 1 \\ 3 & 2 \\ 5 & 6 \end{pmatrix}$$

```
LA =
[ [0,1],
  [3,2],
  [5,6] ]
```

# Double iteration: matrices

Element $a_{i,j}$ of matrix A
is encoded as element L[i-1][j-1] of list LA
(supposing that the indexes of A start at 1)
For simplicity: suppose that both start from zero

$$A = \begin{pmatrix} 0 & 1 \\ 3 & 2 \\ 5 & 6 \end{pmatrix}$$

```
LA =
[ [0,1],
  [3,2],
  [5,6] ]
```

# Double iteration: matrices

## Exercise 1:

## Write a function

```
def is_sum_mat(A,B,S):
    '''A, B, S: matrices encoded as lists
verify that S is the  matrix sum of A and B:
for any i and j: S[i][j]=A[i][j]+B[i][j]'''
```

# Double iteration: matrices

Exercise 2:

Write a function

```
def sum_mat(A,B):
    '''A, B: matrices encoded as lists
```

*return a new list S,* the matrix sum of A and B, as a list of lists'''

# Double iteration: matrices

Exercises 3-4-5:

(1) Function which computes and returns the *transpose* of a matrix, taken as a parameter

(2) Function which takes two *lists* (non matrices!) of the same length, and computes and returns the *scalar product of the two lists*

(3) Function which returns the *product of two matrices* A and B, given as parameters (you may assume they are in the correct format).

Hint: you may use the transpose to make the scalar product between the rows of A and the rows of the transpose of B

# Recursion

Functions may call functions in their body

Hence a function may call *itself* in its body

In this case, we say that that function is *recursive*

*The name of a function is in scope within the function*


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# recursion: ingredients

base case
recursive case

termination :
the recursive case "recurs" on a *simpler* case

*simpler* :
closer to the base case (whatever this means)

As in the case of `while`:
no linguistic guarantee that this is the case
It is the responsability of the programmer

# Recursion: a form of repetition

```
def printrec(n):
    '''print "OK" n times,
     then print "done!" '''
    if n==0:
        print(done!')
    else:
        print('OK')
        printrec(n-1)
```

# Recursion: factorial

n! =n*(n-1)*…*2*1

0!=1

n! = n*(n-1)!

3! = 3*2!=3*(2*1!)=3*(2*(1*0!))=3*(2*(1*1))=6

```
def fact(n):
    if n==0:
        return 1
    else:
        return n*fact(n-1)
```

# Recursion: factorial

```
def fact(n):
    if n==0:
        return 1
    else:
        return n*fact(n-1)
```

Trace the execution of fact(3) on pythontutor

Multiple active frames on the stack

Multiple instances of the local names

# recursion: example

Verify if a string is palindromic


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# recursion

A general theorem of the theory of computation:

Any computation that is expressible through iteration is also expressible through recursion. And vice versa.

# recursion: home exercises

Write recursive functions for the following problems. Also, write an iterative solution.

1. sum of a sequence of integers
2. linear search: given an unordered sequence S and an object el, return True iff el is an element of S
3. Your recursive solution to (2) probably uses slices, that is *copies* of portions of S.
   Write a completely *in place* version (no slices).

   Hint: write a recursive auxiliary function
   `ric_lin_aux(S,index,el)`
   which returns `True` iff `el` is element of the portion of `S` from index to the end.
   Then the real function `ric_lin(S,el)` is simply an initial call to
   `ric_lin_aux(S,0,el)`

file search-rec.py